

NK Embedding of the Two-Maturity FTPL Model with Long-End Captive Demand: Full System and Log-Linearization

Lucas Leturia

Master's Dissertation, Pontificia Universidad Católica de Chile

Entrega III · May 2026

1 Overview

This note embeds the two-maturity FTPL model with long-end captive demand of §3 into a minimal New-Keynesian framework with sticky prices and a passive Taylor rule. The result is a fully dynamic stochastic system whose log-linearization around the deterministic steady state can be solved with standard methods (Klein, gensys, Dynare) to produce impulse-response functions.

The minimum additions to the original model are:

1. A continuum of monopolistically competitive firms producing differentiated varieties with Calvo price stickiness, yielding a New-Keynesian Phillips curve.
2. A central bank following a passive Taylor rule on the nominal short rate $\hat{i}_t$, with feedback coefficient $\phi_\pi < 1$.
3. A consumption Euler equation derived from the household problem in deviations form, which under sticky prices becomes the IS equation linking output to expected real interest rates.

The captive-demand block (mandate $Q_t^L b_t^L = \lambda_t^L V_t^{\text{liab}}$, wedge ω_t , rent R_t) is preserved exactly as in §3. The monetary policy is passive ($\phi_\pi < 1$) because we operate in the fiscal-dominance regime: the price level is pinned by the augmented FTPL identity, not by the Taylor rule.

2 Full Nonlinear System

We collect the equilibrium conditions. All variables are real except where noted; nominal objects are P_t , B_t^S , B_t^L , Q_t^S , Q_t^L . Time subscripts on parameters indicate AR(1) exogenous processes.

Household block

The representative household maximizes

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\sigma}}{1-\sigma}$$

subject to the period budget constraint

$$c_t + Q_t^S b_t^S = y_t - \tau_t + \frac{b_{t-1}^S}{\pi_t}.$$

Short-bond Euler equation:

$$Q_t^S = \mathbb{E}_t \left[\frac{M_{t+1}}{\pi_{t+1}} \right], \quad M_{t+1} \equiv \beta \left(\frac{c_{t+1}}{c_t} \right)^{-\sigma}. \quad (1)$$

Letting $i_t \equiv 1/Q_t^S - 1$ denote the net nominal short rate, this rearranges to the standard nominal Euler equation:

$$\frac{1}{1+i_t} = \beta \mathbb{E}_t \left[\left(\frac{c_{t+1}}{c_t} \right)^{-\sigma} \frac{1}{\pi_{t+1}} \right]. \quad (2)$$

Production block: Calvo Phillips curve

A continuum of monopolistically competitive firms indexed by $i \in [0, 1]$ produces differentiated varieties using labor. The final good is a CES aggregate

$$y_t = \left(\int_0^1 y_{it}^{(\varepsilon-1)/\varepsilon} di \right)^{\varepsilon/(\varepsilon-1)}, \quad \varepsilon > 1.$$

Firms can re-optimize their price each period with probability $1 - \theta$ (Calvo, 1983), so a fraction θ keeps last period's price. The standard derivation (see Galí, 2015, Ch. 3) gives, after log-linearization, the New-Keynesian Phillips curve relating inflation to the real marginal cost (which under our preferences equals the output gap up to a constant):

$$\pi_t = \beta \mathbb{E}_t[\pi_{t+1}] + \kappa (y_t - y_t^n), \quad (3)$$

where y_t^n is the flexible-price level of output, and the slope $\kappa = (1 - \theta)(1 - \beta\theta)/\theta \cdot (\sigma + \varphi)$ depends on the Calvo parameter θ and the labor-supply elasticity φ (when labor is supplied elastically). With endowment-economy preferences and no labor margin, κ reduces to $(1 - \theta)(1 - \beta\theta)\sigma/\theta$. Equation (3) is already approximately log-linear; its exact form is the standard Calvo-Yun derivation.

Long-bond block

The long bond is a nominal geometric perpetuity paying coupons $1, \delta, \delta^2, \dots$ at $t + 1, t + 2, \dots$. Its fundamental price is

$$Q_t^{L,\text{fund}} = \mathbb{E}_t \left[\frac{M_{t+1}(1 + \delta Q_{t+1}^L)}{\pi_{t+1}} \right]. \quad (4)$$

The captive-market clearing condition is

$$Q_t^L b_t^L = \lambda_t^L V_t^{\text{liab}}. \quad (5)$$

The wedge ω_t is defined by

$$1 + \omega_t \equiv \frac{Q_t^L}{Q_t^{L,\text{fund}}}, \quad (6)$$

and the per-period rent is

$$R_t = \frac{\omega_t}{1 + \omega_t} Q_t^L b_t^L. \quad (7)$$

Government block

The government's real flow budget constraint is

$$\frac{b_{t-1}^S + (1 + \delta Q_t^L) b_{t-1}^L}{\pi_t} = s_t + Q_t^S b_t^S + Q_t^L b_t^L, \quad (8)$$

with $s_t = \tau_t - g_t$. The maturity composition is exogenous:

$$b_t^L = \chi_t (b_t^S + b_t^L). \quad (9)$$

Define real market value of debt

$$v_t \equiv \frac{b_{t-1}^S + (1 + \delta Q_t^L) b_{t-1}^L}{\pi_t}. \quad (10)$$

The augmented FTPL identity (derived in §3) is

$$v_t = s_t + R_t + \mathbb{E}_t[M_{t+1} v_{t+1}]. \quad (11)$$

Monetary policy

A passive Taylor rule governs the nominal short rate:

$$i_t = i^* + \phi_\pi (\pi_t - \pi^*) + u_t^i, \quad \phi_\pi < 1, \quad (12)$$

where $i^* = 1/\beta - 1 + (\pi^* - 1)$ is the steady-state nominal rate consistent with the inflation target π^* , and u_t^i is a monetary policy shock. Passivity is essential: under $\phi_\pi > 1$ (active Taylor), the price level is pinned by monetary policy and the FTPL channel is absent. Under $\phi_\pi < 1$, monetary policy accommodates inflation and the price level is determined by the augmented FTPL identity (11).

Resource constraint

$$c_t + g_t = y_t. \quad (13)$$

Exogenous processes

$$\log \lambda_t^L = (1 - \rho_\lambda) \log \bar{\lambda}^L + \rho_\lambda \log \lambda_{t-1}^L + \varepsilon_t^\lambda, \quad \varepsilon_t^\lambda \sim \text{i.i.d. } \mathcal{N}(0, \sigma_\lambda^2), \quad (14)$$

$$\log \chi_t = (1 - \rho_\chi) \log \bar{\chi} + \rho_\chi \log \chi_{t-1} + \varepsilon_t^\chi, \quad \varepsilon_t^\chi \sim \text{i.i.d. } \mathcal{N}(0, \sigma_\chi^2), \quad (15)$$

$$\log V_t^{\text{liab}} = (1 - \rho_V) \log \bar{V}^{\text{liab}} + \rho_V \log V_{t-1}^{\text{liab}} + \varepsilon_t^V, \quad \varepsilon_t^V \sim \text{i.i.d. } \mathcal{N}(0, \sigma_V^2), \quad (16)$$

$$\log g_t = (1 - \rho_g) \log \bar{g} + \rho_g \log g_{t-1} + \varepsilon_t^g, \quad \varepsilon_t^g \sim \text{i.i.d. } \mathcal{N}(0, \sigma_g^2), \quad (17)$$

$$\log \tau_t = (1 - \rho_\tau) \log \bar{\tau} + \rho_\tau \log \tau_{t-1} + \varepsilon_t^\tau, \quad \varepsilon_t^\tau \sim \text{i.i.d. } \mathcal{N}(0, \sigma_\tau^2), \quad (18)$$

$$u_t^i = \rho_i u_{t-1}^i + \varepsilon_t^i, \quad \varepsilon_t^i \sim \text{i.i.d. } \mathcal{N}(0, \sigma_i^2). \quad (19)$$

We treat output y_t as exogenous in the headline specification (endowment economy with the Phillips curve referring to the gap to a flexible-price natural output; this preserves the original model's tractability and keeps the focus on the fiscal-monetary block). An extension that endogenizes y_t via a labor-leisure choice is straightforward but not needed for the headline IRFs.

3 Deterministic Steady State

In the deterministic steady state, all variables are constant, expectations equal realizations, and shocks are zero. We index steady-state values without time subscripts. The steady-state inflation rate $\bar{\pi} \equiv \pi^*$ is set by the central bank's target.

Steady-state values

- Real SDF: $M = \beta$ (since $c_{t+1}/c_t = 1$).
- Short-bond price: $Q^S = \beta/\bar{\pi}$.
- Net nominal short rate: $i^* = \bar{\pi}/\beta - 1$.
- Long-bond fundamental price: $Q^{L,\text{fund}} = (\beta/\bar{\pi})/(1 - \beta\delta/\bar{\pi})$.
- Long-bond price: $Q^L = (1 + \omega)Q^{L,\text{fund}}$.
- Rent: $R = [\omega/(1 + \omega)] \bar{\lambda}^L \bar{V}^{\text{liab}}$.
- Real debt market value: $v = [\bar{b}^S + (1 + \delta Q^L)\bar{b}^L]/\bar{\pi}$.
- Augmented FTPL: $v = (\bar{s} + R)/(1 - \beta)$.
- Output: $\bar{y}$ (normalized).
- Consumption: $\bar{c} = \bar{y} - \bar{g}$.

The wedge ω and stock b^L are jointly determined by the mandate equation $Q^L b^L = \bar{\lambda}^L \bar{V}^{\text{liab}}$ together with the wedge definition (6), which gives the closed-form expression derived in §3 of the dissertation.

4 Log-Linearization

We log-linearize each equilibrium condition around the steady state. For a variable x_t , define $\hat{x}_t \equiv \log x_t - \log \bar{x}$ (the log-deviation from steady state). For variables that can be negative in level (e.g., the primary surplus s_t), we use deviations from steady state divided by the relevant scale: $\hat{s}_t \equiv (s_t - \bar{s})/\bar{y}$.

(i) Euler equation $\Rightarrow$ IS equation

Take the Euler (2) in logs and expand around steady state. Using $\log(1 + i_t) \approx i_t - i^*$ (call this $\hat{i}_t$ in deviation form), $\log \beta = \log(\beta/\bar{\pi}) + \log \bar{\pi}$, and the consumption-growth term to first order:

$$\boxed{\hat{c}_t = \mathbb{E}_t[\hat{c}_{t+1}] - \frac{1}{\sigma}(\hat{i}_t - \mathbb{E}_t[\hat{\pi}_{t+1}])} \quad (20)$$

Using the resource constraint (13) log-linearized as $\bar{c} \hat{c}_t = \bar{y} \hat{y}_t - \bar{g} \hat{g}_t$, this rewrites equivalently in terms of the output gap. We use the consumption form for IRF interpretation.

(ii) Phillips curve

Equation (3) is already in approximately log-linear form. Using $\hat{\pi}_t = \log(\pi_t/\bar{\pi})$ and the output gap $\hat{y}_t - \hat{y}_t^n$, where $\hat{y}_t^n$ is the log-deviation of natural output (zero in our endowment-economy setup with constant productivity):

$$\boxed{\hat{\pi}_t = \beta \mathbb{E}_t[\hat{\pi}_{t+1}] + \kappa \hat{y}_t} \quad (21)$$

with $\kappa = (1 - \theta)(1 - \beta\theta)\sigma/\theta$ in the endowment-economy specification.

(iii) Short-bond pricing

From (2): $\hat{i}_t = \hat{r}_t + \mathbb{E}_t[\hat{\pi}_{t+1}]$ where $\hat{r}_t \equiv \hat{i}_t - \mathbb{E}_t[\hat{\pi}_{t+1}]$ is the ex-ante real rate. Equivalently $\hat{Q}_t^S = -\hat{i}_t/(1+i^*)$, which at first order is $\hat{Q}_t^S \approx -\hat{i}_t \cdot (\beta/\bar{\pi})$, but it is cleaner to work directly with $\hat{i}_t$ as a state variable rather than $\hat{Q}_t^S$. We do so below.

(iv) Long-bond fundamental pricing

The nonlinear equation is $Q_t^{L,\text{fund}} = \mathbb{E}_t[M_{t+1}(1 + \delta Q_{t+1}^L)/\pi_{t+1}]$. Taking logs and expanding:

$$\log Q_t^{L,\text{fund}} = \log \mathbb{E}_t \left[\beta \left(\frac{c_{t+1}}{c_t} \right)^{-\sigma} \frac{1 + \delta Q_{t+1}^L}{\pi_{t+1}} \right].$$

Standard first-order approximation (log-normal expectation kernel collapses to a linear function of conditional means):

$$\hat{Q}_t^{L,\text{fund}} = -\sigma(\mathbb{E}_t \hat{c}_{t+1} - \hat{c}_t) - \mathbb{E}_t \hat{\pi}_{t+1} + \frac{\delta Q^L}{1 + \delta Q^L} \mathbb{E}_t \hat{Q}_{t+1}^L. \quad (22)$$

Define the coefficient $\eta \equiv \delta Q^L/(1 + \delta Q^L)$. This is a key reduced-form parameter: at the calibration of §3.4 ($Q^L \approx 22$, $\delta \approx 0.96$), $\eta \approx 0.95$, very close to one, reflecting the long duration of the bond.

(v) Long-bond market clearing

Take logs of $Q_t^L b_t^L = \lambda_t^L V_t^{\text{liab}}$:

$$\boxed{\hat{Q}_t^L + \hat{b}_t^L = \hat{\lambda}_t^L + \hat{V}_t^{\text{liab}}} \quad (23)$$

(vi) Wedge

From $1 + \omega_t = Q_t^L / Q_t^{L,\text{fund}}$: take logs and expand,

$$\log(1 + \omega_t) = \log Q_t^L - \log Q_t^{L,\text{fund}}.$$

At first order, $\log(1 + \omega_t) \approx \log(1 + \bar{\omega}) + \frac{1}{1 + \bar{\omega}}(\omega_t - \bar{\omega})$. Defining $\hat{\omega}_t \equiv (\omega_t - \bar{\omega})/(1 + \bar{\omega})$ (a natural rescaling),

$$\hat{\omega}_t = \hat{Q}_t^L - \hat{Q}_t^{L,\text{fund}}. \quad (24)$$

(vii) Rent

From $R_t = [\omega_t/(1 + \omega_t)]Q_t^L b_t^L$: take logs,

$$\log R_t = \log \omega_t - \log(1 + \omega_t) + \log Q_t^L + \log b_t^L.$$

Define $\hat{R}_t \equiv (R_t - \bar{R})/\bar{R}$ (proportional deviation). At first order, using $\omega/(1 + \omega) = R/(\bar{\lambda}^L \bar{V}^{\text{liab}})$ in steady state:

$$\hat{R}_t = \frac{1}{\bar{\omega}(1 + \bar{\omega})} [\hat{Q}_t^L - \hat{Q}_t^{L,\text{fund}}] + \hat{\lambda}_t^L + \hat{V}_t^{\text{liab}}, \quad (25)$$

where we have used (23) to substitute for $\hat{Q}_t^L + \hat{b}_t^L$. Equivalently, using (24): $\hat{R}_t = (1/\bar{\omega})\hat{\omega}_t + \hat{\lambda}_t^L + \hat{V}_t^{\text{liab}}$.

(viii) Government flow budget constraint

Multiply (8) through and log-linearize. Define shares at steady state:

$$\alpha_{bS} \equiv \frac{\bar{b}^S}{\bar{b}^S + (1 + \delta Q^L)\bar{b}^L/\bar{\pi}}, \quad \alpha_{bL} \equiv 1 - \alpha_{bS}.$$

The log-linearized flow constraint becomes a relation among $\hat{b}_{t-1}^S, \hat{b}_{t-1}^L, \hat{Q}_t^L, \hat{\pi}_t, \hat{b}_t^S, \hat{b}_t^L$ and $\hat{s}_t$. The bookkeeping is tedious but mechanical; we collect it as

$$\begin{aligned} \alpha_{bS}(\hat{b}_{t-1}^S - \hat{\pi}_t) + \alpha_{bL}\left(\hat{b}_{t-1}^L + \frac{\delta Q^L}{1 + \delta Q^L}\hat{Q}_t^L - \hat{\pi}_t\right) \\ = (1 - \beta)\hat{s}_t + \beta(\hat{b}_t^S \cdot \frac{Q^S \bar{b}^S}{v\bar{\pi}} + \hat{b}_t^L \cdot \frac{Q^L \bar{b}^L}{v\bar{\pi}}) + (\text{lower-order}), \end{aligned} \quad (26)$$

where the precise scaling depends on the steady-state ratios. For implementation in Dynare it is easier to work directly with the recursive form of the FTPL identity (11), which we log-linearize separately below.

(ix) Real debt market value

From (10) and the steady state $v = [\bar{b}^S + (1 + \delta Q^L)\bar{b}^L]/\bar{\pi}$, define shares

$$w_S \equiv \frac{\bar{b}^S/\bar{\pi}}{v}, \quad w_L \equiv \frac{(1 + \delta Q^L)\bar{b}^L/\bar{\pi}}{v}, \quad w_S + w_L = 1.$$

Log-linearizing v_t :

$$\hat{v}_t = w_S(\hat{b}_{t-1}^S - \hat{\pi}_t) + w_L\left(\hat{b}_{t-1}^L + \frac{\delta Q^L}{1 + \delta Q^L}\hat{Q}_t^L - \hat{\pi}_t\right). \quad (27)$$

The price-level term enters with the coefficient $-(w_S + w_L) = -1$: a 1% increase in π_t reduces real debt by 1%, holding nominal stocks constant. This is the inflation-reevaluation channel at the heart of the FTPL.

(x) Recursive FTPL identity

Linearize (11): $v_t = s_t + R_t + \mathbb{E}_t[M_{t+1}v_{t+1}]$. In deviations:

$$\boxed{\hat{v}_t = \frac{\bar{s}}{v}\hat{s}_t + \frac{\bar{R}}{v}\hat{R}_t + \beta \mathbb{E}_t[\hat{v}_{t+1} - \sigma(\hat{c}_{t+1} - \hat{c}_t)]} \quad (28)$$

where the $\sigma(\hat{c}_{t+1} - \hat{c}_t)$ term arises from log-linearizing M_{t+1} . Using the IS equation (20), $\sigma(\mathbb{E}_t\hat{c}_{t+1} - \hat{c}_t) = \hat{i}_t - \mathbb{E}_t\hat{\pi}_{t+1}$, so the discount-factor term can be rewritten as $\beta \mathbb{E}_t\hat{v}_{t+1} - \beta(\hat{i}_t - \mathbb{E}_t\hat{\pi}_{t+1})$. Substituting:

$$\hat{v}_t = \frac{\bar{s}}{v}\hat{s}_t + \frac{\bar{R}}{v}\hat{R}_t + \beta \mathbb{E}_t[\hat{v}_{t+1}] - \beta(\hat{i}_t - \mathbb{E}_t[\hat{\pi}_{t+1}]). \quad (29)$$

(xi) Taylor rule

Linearize (12):

$$\boxed{\hat{i}_t = \phi_\pi \hat{\pi}_t + u_t^i}, \quad \phi_\pi < 1. \quad (30)$$

(xii) Maturity composition

Linearize (9): with $\bar{\chi} = \bar{b}^L/(\bar{b}^S + \bar{b}^L)$,

$$\hat{b}_t^L = \hat{\chi}_t + (1 - \bar{\chi})(\hat{b}_t^S - \hat{b}_t^L) \cdot \frac{\bar{b}^S}{\bar{b}^S + \bar{b}^L}. \quad (31)$$

Equivalently, $\hat{b}_t^L - \hat{b}_t^S = \hat{\chi}_t/[\bar{\chi}(1 - \bar{\chi})]$ to first order.

(xiii) Resource constraint

$$\bar{c}\hat{c}_t + \bar{g}\hat{g}_t = \bar{y}\hat{y}_t. \quad (32)$$

(xiv) Primary surplus

The primary surplus is composed of taxes net of spending, $s_t = \tau_t - g_t$, both exogenous AR(1) processes:

$$\hat{s}_t = \frac{\bar{\tau}}{\bar{y}}\hat{\tau}_t - \frac{\bar{g}}{\bar{y}}\hat{g}_t. \quad (33)$$

5 Summary of the Log-Linearized System

#	Equation	Role
1	$\hat{c}_t = \mathbb{E}_t[\hat{c}_{t+1}] - \frac{1}{\sigma}(\hat{i}_t - \mathbb{E}_t[\hat{\pi}_{t+1}])$	IS / consumption Euler
2	$\hat{\pi}_t = \beta\mathbb{E}_t[\hat{\pi}_{t+1}] + \kappa\hat{y}_t$	New-Keynesian Phillips curve
3	$\hat{i}_t = \phi_\pi\hat{\pi}_t + u_t^i$	Passive Taylor rule ($\phi_\pi < 1$)
4	$\hat{Q}_t^{L,\text{fund}} = -\sigma(\mathbb{E}_t\hat{c}_{t+1} - \hat{c}_t) - \mathbb{E}_t\hat{\pi}_{t+1} + \eta\mathbb{E}_t\hat{Q}_{t+1}^L$	Long-bond fundamental price
5	$\hat{Q}_t^L + \hat{b}_t^L = \hat{\lambda}_t^L + \hat{V}_t^{\text{liab}}$	Captive market clearing
6	$\hat{\omega}_t = \hat{Q}_t^L - \hat{Q}_t^{L,\text{fund}}$	Wedge definition
7	$\hat{R}_t = \hat{\omega}_t/\bar{\omega} + \hat{\lambda}_t^L + \hat{V}_t^{\text{liab}}$	Rent
8	$\hat{v}_t = w_S(\hat{b}_{t-1}^S - \hat{\pi}_t) + w_L(\hat{b}_{t-1}^L + \eta\hat{Q}_t^L - \hat{\pi}_t)$	Real debt market value
9	$\hat{v}_t = \frac{\bar{s}}{\bar{v}}\hat{s}_t + \frac{\bar{R}}{\bar{v}}\hat{R}_t + \beta\mathbb{E}_t\hat{v}_{t+1} - \beta(\hat{i}_t - \mathbb{E}_t\hat{\pi}_{t+1})$	FTPL identity
10	$\hat{b}_t^L - \hat{b}_t^S = \hat{\chi}_t/[\bar{\chi}(1 - \bar{\chi})]$	Maturity composition
11	$\bar{c}\hat{c}_t + \bar{g}\hat{g}_t = \bar{y}\hat{y}_t$	Resource constraint
12	$\hat{s}_t = (\bar{\tau}/\bar{y})\hat{\tau}_t - (\bar{g}/\bar{y})\hat{g}_t$	Primary surplus
13–18	AR(1) processes for $\lambda_t^L, \chi_t, V_t^{\text{liab}}, g_t, \tau_t, u_t^i$	Exogenous block

Endogenous variables (12): $\hat{c}_t, \hat{y}_t, \hat{\pi}_t, \hat{i}_t, \hat{Q}_t^L, \hat{Q}_t^{L,\text{fund}}, \hat{b}_t^S, \hat{b}_t^L, \hat{\omega}_t, \hat{R}_t, \hat{v}_t, \hat{s}_t$.

Exogenous variables (6): $\hat{\lambda}_t^L, \hat{\chi}_t, \hat{V}_t^{\text{liab}}, \hat{g}_t, \hat{\tau}_t, u_t^i$.

Structural shocks (6): $\varepsilon_t^\lambda, \varepsilon_t^\chi, \varepsilon_t^V, \varepsilon_t^g, \varepsilon_t^\tau, \varepsilon_t^i$.

The system has 12 dynamic equations in 12 endogenous variables plus 6 exogenous AR(1) processes. With $\phi_\pi < 1$ (passive monetary policy), the Blanchard-Kahn conditions require the FTPL identity (29) to be the equation that pins down inflation expectations — this is the formal mechanism by which fiscal dominance produces a unique price-level equilibrium.

6 Determinacy: Why Passive Monetary Policy Is Essential

Under active monetary policy ($\phi_\pi > 1$, the Taylor principle), the standard NK three-equation system (IS, NKPC, Taylor) has a unique stable equilibrium with inflation pinned by the Taylor rule. Adding the FTPL identity (29) *overdetermines* the system: any path of π_t that satisfies the Taylor-rule equilibrium generically violates the FTPL identity. To restore consistency, either fiscal policy must adjust passively (the standard Ricardian closure) or monetary policy must be passive.

Under passive monetary policy ($\phi_\pi < 1$), the NK three-equation system has multiple equilibria (the IS-NKPC pair plus the Taylor rule fails to pin down a unique price level). The FTPL identity selects among these by requiring real debt to equal the present value of surpluses plus rents. This is the formal expression of fiscal dominance.

In our model, the captive-demand block enriches this story: the FTPL identity now contains the rent term R_t , which depends on the policy variable λ_t^L . A shock to λ_t^L alters the rent path, which alters the implied PDV of debt backing, which forces a revaluation of π_t through equation (29). *This is the mechanism by which financial repression policy affects the price level.*

7 Steady-State Coefficients for the Calibration of §3.4

For numerical implementation, we collect the reduced-form coefficients implied by the UK calibration ($\beta = 0.99$, $\delta = 0.96$, $\bar{\omega} = 0.121$, primary deficit -3% GDP, etc.):

Symbol	Definition	Value
η	$\delta Q^L / (1 + \delta Q^L)$	≈ 0.95
w_S	$(\bar{b}^S / \bar{\pi}) / v$	≈ 0.63
w_L	$(1 + \delta Q^L) \bar{b}^L / (v \bar{\pi})$	≈ 0.37
$\bar{s} / v$	primary balance / debt market value	≈ -0.03
$\bar{R} / v$	rent / debt market value	≈ 0.04
$1 / \bar{\omega}$	inverse wedge	≈ 8.26
β	discount factor	0.99
σ	CRRA	1 (convention)
κ	NKPC slope (with $\theta = 0.75$, $\sigma = 1$)	≈ 0.086
ϕ_π	Taylor coefficient (passive)	0.5 (suggested)

8 Notes Toward the Dynare Implementation

The Dynare script will declare 12 endogenous variables, 6 exogenous AR(1) variables and 6 structural shocks. The model block transcribes equations 1–12 above. The steady state can be supplied as an explicit `steady_state_model` block using the closed-form expressions of §3.4. Order of operations:

1. Declare variables and parameters; set parameter values from the calibration table.
2. Write the model block transcribing equations 1–12 of the summary table.
3. Declare AR(1) exogenous processes (equations 13–18).

4. Provide an explicit steady-state block.
5. Run `check`; and `stoch_simul` to confirm Blanchard-Kahn conditions are satisfied.
6. Generate IRFs to the policy shock ε_t^λ over a 40-quarter horizon, focusing on $\hat{\pi}_t, \hat{Q}_t^L, \hat{R}_t, \hat{v}_t$.
7. Report sensitivity of $\hat{\pi}_t$ at impact and at horizons $h = 4, 8, 20$ to alternative values of ϕ_π and ρ_λ .

The headline IRF to bring to the advisor: *response of inflation $\hat{\pi}_t$ to a one-standard-deviation positive shock in the regulatory intensity $\hat{\lambda}_t^L$ under passive monetary policy ($\phi_\pi = 0.5$).* The model predicts a negative response on impact (a tightening of repression substitutes for inflation as the equilibrating margin) that decays at the rate ρ_λ .

9 Caveats

This embedding makes several minimalist choices that should be flagged:

1. **No labor margin.** The Phillips curve refers to a notional output gap; under endowment-economy preferences this collapses to a consumption gap. A fully fledged version would add a labor-leisure choice and the standard wage equation. This is a one-line addition but is not done here.
2. **σ and κ identification.** σ becomes identified once dynamic moments are available; κ requires a price-stickiness parameter θ which the steady-state calibration cannot pin down. Both are set to standard textbook values for the headline IRFs.
3. **Approximate vs. exact η .** The coefficient $\eta = \delta Q^L / (1 + \delta Q^L)$ depends on the steady-state value of ω , which is itself an output of the calibration. This circularity is resolved by computing the steady state first (using §3.4), then plugging $\bar{\omega}$ into the log-linearized coefficients.
4. **Maturity-composition rule.** Equation (31) assumes the government can rebalance the maturity composition each period. An alternative would be to specify exogenous bond issuance, with χ_t determined endogenously from the resulting stock dynamics. The headline IRFs do not depend strongly on this choice.